Three Worlds of 2041

How AI could reshape our industry, our customers and our planet over the next 15 years

THREE SCENARIOS, NOT THREE PREDICTIONS

On 6 April 2026, four people watched Earth set behind the Moon. Fifteen years from now, the Moon may be a place where people go to work. This is not a dream. It is three futures we can already see coming, and the choices that decide which one we get.

IN THIS BRIEFING today's signals, our method, three worlds of 2041, the odds, and what to do from Monday.

Mohammad
Senior Futures Consultant, Em, Oli & Timmonds
Part 1 | The Creative Solution

Earth sets behind the Moon, photographed by the Artemis II crew, 6 April 2026. Image: NASA

One story, fifteen steps

Our brief: imagine the world in 15 years' time, considering AI's impact on our industry and on the whole world, and decide what E,O&T should build now without betting on one forecast. The story uses an adapted, branching Story Spine: the three "Because of that" steps are alternative worlds. Facts and projections are tagged; the 2041 worlds are imagined scenarios.

3 EVERY DAY
The future is leaking in
TOOL 2 of the 4 lenses

4 BUT ONE DAY
The Futures Engine
TOOL Seven creative tools

5 BECAUSE OF THAT (1)
Symbiosis 2041
TOOL Moonshot, upward What if

6 BECAUSE OF THAT (2)
Overheat 2041
TOOL Downward What if

7 BECAUSE OF THAT (3)
Patchwork 2041
TOOL Tick-Tock

8 ZOOM IN (1)
From your door to orbit
TOOL Moonshot, exaptation

9 ZOOM IN (2)
The new gold
TOOL Tick-Tock, all 4 lenses

10 UNTIL FINALLY
Which branch are we on?
TOOL Possibility tree

11 UNTIL FINALLY: WATCH
Lights that change action
TOOL Contingency planning

12 EVER SINCE: RESPOND
Sell better decisions
TOOL Decision Passport

13 STORY IN ONE WORLD
Omar in eight steps
TOOL Story Spine, 5 Cs

14 AND EVER SINCE THEN
Back to Monday morning
TOOL Working backwards

15 EVER SINCE: FIRST TEST
The first 90 days
TOOL Working backwards, gates

16 EVER SINCE: KEEP CREATING
A creative routine
TOOL Safety, Octalysis, stimuli

17 THE CARRY-OUT MESSAGE
The Moon is closer than ever
TOOL 5 Cs of storytelling

SCOPE Industry: strategic consulting | Clients: energy, logistics and digital services | World: work, daily life, infrastructure and the space frontier

NEXT we start with the facts: four signals already visible in 2026.

The future is already leaking in

Four signals we can measure in 2026, found with two of the 4 lenses of innovation: harnessing trends and challenging orthodoxies. Every world grows out of them.

2026 — 2041

415 TWh

used by data centres in 2024, about 1.5% of world electricity, with about 945 TWh by 2030 as a base-case projection. (IEA)

Energy becomes AI's ceiling.

170M new jobs

created and 92M displaced by 2030 across major economic and technological trends, including AI; 39% of core skills change. (WEF)

Skills, not jobs, become the scarce asset.

54,000

debris objects larger than 10 cm circle Earth, plus about 1.2 million between 1 and 10 cm. (ESA)

Orbit is a finite resource we can pollute.

April 2026

Artemis II flies four astronauts around the Moon, the first crewed lunar flight since 1972. (NASA)

Space moves from exploration to economy.

Challenging orthodoxies. The old belief "AI is software" no longer holds. AI is an energy, materials and orbit industry, and that is exactly where our three worlds split apart.

The E,O&T Futures Engine

How the signals become worlds. Seven tools work like gears, and each produces a visible output on the slides shown. We first ran a controlled What if (keep AI access fixed, change only energy headroom), then combined both uncertainties into three worlds.

1 **Harness trends**
Spot what is already moving
SLIDES 3, 9

2 **Counterfactual What if**
Change one driver, upward and downward
SLIDES 5, 6

3 **Possibility tree**
Branch it and put odds on every branch
SLIDES 10

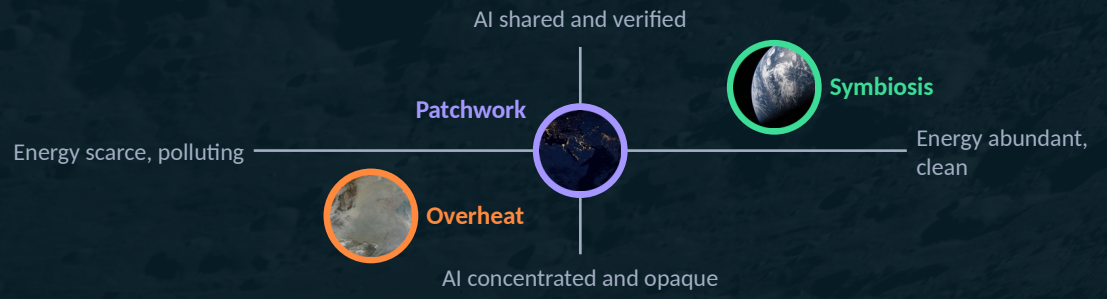
4 **Tick-Tock**
Flip each assumption into its opposite
SLIDES 7, 9

5 **Moonshot 10x**
Push the best branch to its boldest form
SLIDES 5, 8

6 **Hard empathy**
Live one day inside each world, in 5 steps
SLIDES 5 to 7, 13

7 **Working backwards**
Plan right to left, from 2041 to today
SLIDES 14, 15

Two critical uncertainties set the axes



AI "bad copying"

We asked AI for dozens of variants of each world and kept the strange ones. Its mistakes became raw material.

Exaptation

Borrowed from air traffic control (one network, many vehicles) and multiplayer game design (rules that shape behaviour).

NEXT the engine's first output: the best world we can imagine.

WORLD 1 OF 3

SYMBIOSIS 2041

STORY SPINE **Because of that (1)**

Moonshot thinking: the upward What if

Shared, verified AI with abundant clean energy

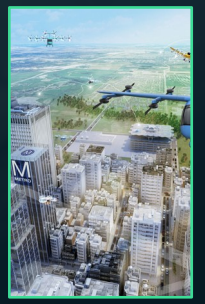
Robots on the Moon (NASA concept)

A future we want to be part of

The best case. We push the engine's upper branch to a moonshot, then live one day inside it through hard empathy.

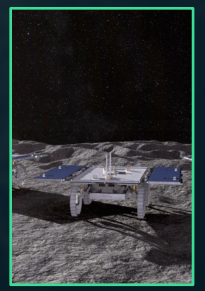
Mirror Cities

Customers try a flat or a lunar hotel in a live digital twin before buying. White-hat Octalysis: each purchase plants a real tree.



Life and transport

Electric pods and air taxis on one AI network, borrowed from air traffic control. Car parks become parks.



Space and the Moon

A crewed base at the lunar south pole mines ice for fuel. Robots build; the first tourists arrive.

Our industry

AI writes the analysis; consultants own the judgement. E,O&T move: launch a future-design practice.

A DAY IN 2041 | HARD EMPATHY IN 5 STEPS

Layla, 34, grid engineer, Riyadh
Scene: 6 am. My AI agent has already sold my solar surplus and drafted my report.
Feel: free, and strangely idle.
Make sense: the machines now do the routine half of my job.
Worry: what am I still for?
Act: I join the council that designs our city's digital twin.

5 Cs

Character	Conflict	Cure	Change	Carry-out
Layla, and every client like her	AI's hunger for energy	Clean power, efficient chips	Abundance, shared widely	Build the future we want

Still hard here: e-waste, attention overload, disputes over lunar mining.

NEXT every utopia has a shadow. The same engine, run downward.

WORLD 2 OF 3

OVERHEAT 2041

Counterfactual: the downward What if

Concentrated, opaque AI with scarce, polluting energy

Hardware in low orbit (NASA, ISS)

A DAY IN 2041 | HARD EMPATHY IN 5 STEPS

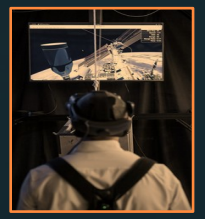
Omar, 41, logistics manager
Scene: in the depot, two delivery alerts contradict each other.
Feel: anxious; the rationing app also limits when I can charge the vans.
Make sense: one of the alerts may be synthetic.
Worry: I could send a driver onto an unsafe route.
Act: I verify both alerts through our data cooperative before dispatch.

Every utopia has a shadow

The honest shadow of World 1. Same engine, downward What if, told as a chain of cause and effect.

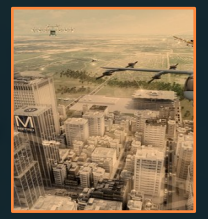
HOW IT HAPPENED, ONE "BECAUSE OF THAT" AT A TIME

1. AI demand outran clean power, so fossil plants kept running.
2. Heat and pollution rose, and power was rationed.
3. A few players owned the models; deepfakes broke trust.
4. A debris cascade took out navigation and broadband satellites.



Customers

The metaverse is an escape; black-hat Octalysis (scarcity, loss) keeps people hooked.



Life and transport

Empty robotaxis gridlock the cities. The rich fly; everyone else waits.

Space and the Moon

Low orbit is a junkyard; the lunar base waits.

Our industry

Reports cost pennies. E,O&T move: verification and resilience services.

5 Cs	Character	Conflict	Cure	Change	Carry-out
	Omar	AI outgrows clean power	None, in time	Trust and orbit lost	Plan for the shadow first

NEXT between heaven and collapse sits the most likely world.

WORLD 3 OF 3

PATCHWORK 2041

Tick-Tock: the middle path

Trust verified inside regional blocs, mixed energy between them

One planet, three rulebooks.

A DAY IN 2041 | HARD EMPATHY IN 5 STEPS

Sara, 29, product designer, Stockholm
Scene: my AI agent is doing my weekly shopping.
Feel: relaxed, until it refuses my favourite app.
Make sense: that app has no trust seal in our bloc.
Worry: my own product could be locked out the same way.
Act: I design one product that passes all three blocs' trust rules.

STORY SPINE Because of that (3)

Neither heaven nor collapse

The middle path, built by flipping two of today's facts with Tick-Tock. Here trust, not technology, decides who wins.

TICK People buy from companies.

TOCK **AI agents buy from AI agents, so firms must design for machine customers.**

TICK Data is free.

TOCK **Data is owned and paid for, like energy.**

Customers

Regional metaverses joined by one digital identity, so trust becomes the product.

Life and transport

Driverless in some regions, manual in others; the best cities blend both.

Space and the Moon

Shared on paper; blocs compete quietly for lunar ice and landing sites.

Our industry

Winners translate between blocs. E,O&T move: a cross-bloc standards advisory.

5 Cs

Character
Sara

Conflict
One world, many rulebooks

Cure
Shared trust seals

Change
Identity is the product

Carry-out
The future is a negotiation

NEXT zoom in: how a food-delivery network becomes an orbital supply line.

From your door to orbit

Borrow the delivery-network logic, not the aircraft: a model that brings dinner today, adapted for orbital logistics, could help feed astronauts in 2041.

2026 OBSERVED

Drones deliver food now

One delivery network passed 2 million commercial drone deliveries in January 2026, with a median flight of 3 minutes (Food On Demand).

TRANSFER: scheduling, tracking, exception handling

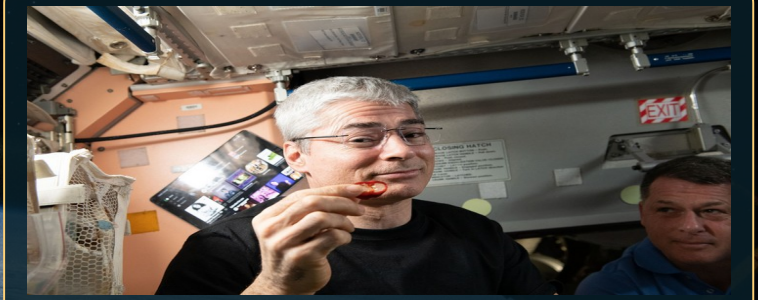
2030s PROPOSED TEST

Autonomy reaches orbit

Robotic freighters already berth at the station using its robotic arm. Next: test AI scheduling with qualified space partners.

GATE: safety, communication delay, payload limits

2041 MOONSHOT



Feeding the crew in space

Each astronaut needs about 1.7 kg of food a day; fresh produce arrives every three months (NASA). Imagine cargo and orbital crops supporting a lunar outpost.

10x AMBITION: on-demand, not occasional, resupply

E,O&T test. Start with a logistics simulation and a partner review, not launch infrastructure. For clients in logistics and food, the last mile becomes the last orbit.

The new gold

TICK Gold is the treasure. **TOCK** **Water, air, sand and plants are.**

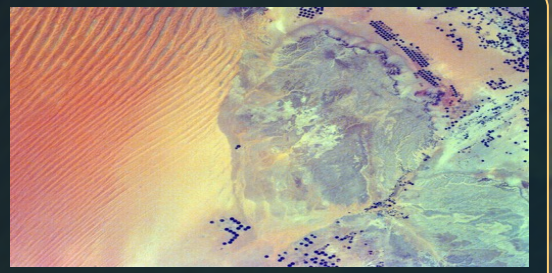
In 2041 value may sit in what keeps systems alive. Each card applies one of the 4 lenses of innovation and ends with an E,O&T opportunity.

Water

Training GPT-3 was estimated to evaporate about 700,000 litres of fresh water (Li et al.). The space station recycles 98% of its water (NASA).

LEVERAGE RESOURCES

Adapt closed-loop thinking to a client water audit.



Sand

Sand and gravel use is about 50 billion tonnes a year, 17 kg per person per day (UNEP). Desert sand is too round for concrete.

CHALLENGE ORTHODOXIES

Replace "abundant means disposable" with reuse by design.

Air

Lunar soil is 40 to 45% oxygen by weight (ESA), but it is chemically bound and not breathable until extracted.

HARNESS TRENDS

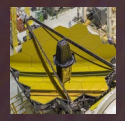
Track extraction tests, energy needs and readiness.

Plants

Lettuce and chile peppers grow on the station (NASA): crop research, not yet a complete food system.

UNDERSTAND NEEDS

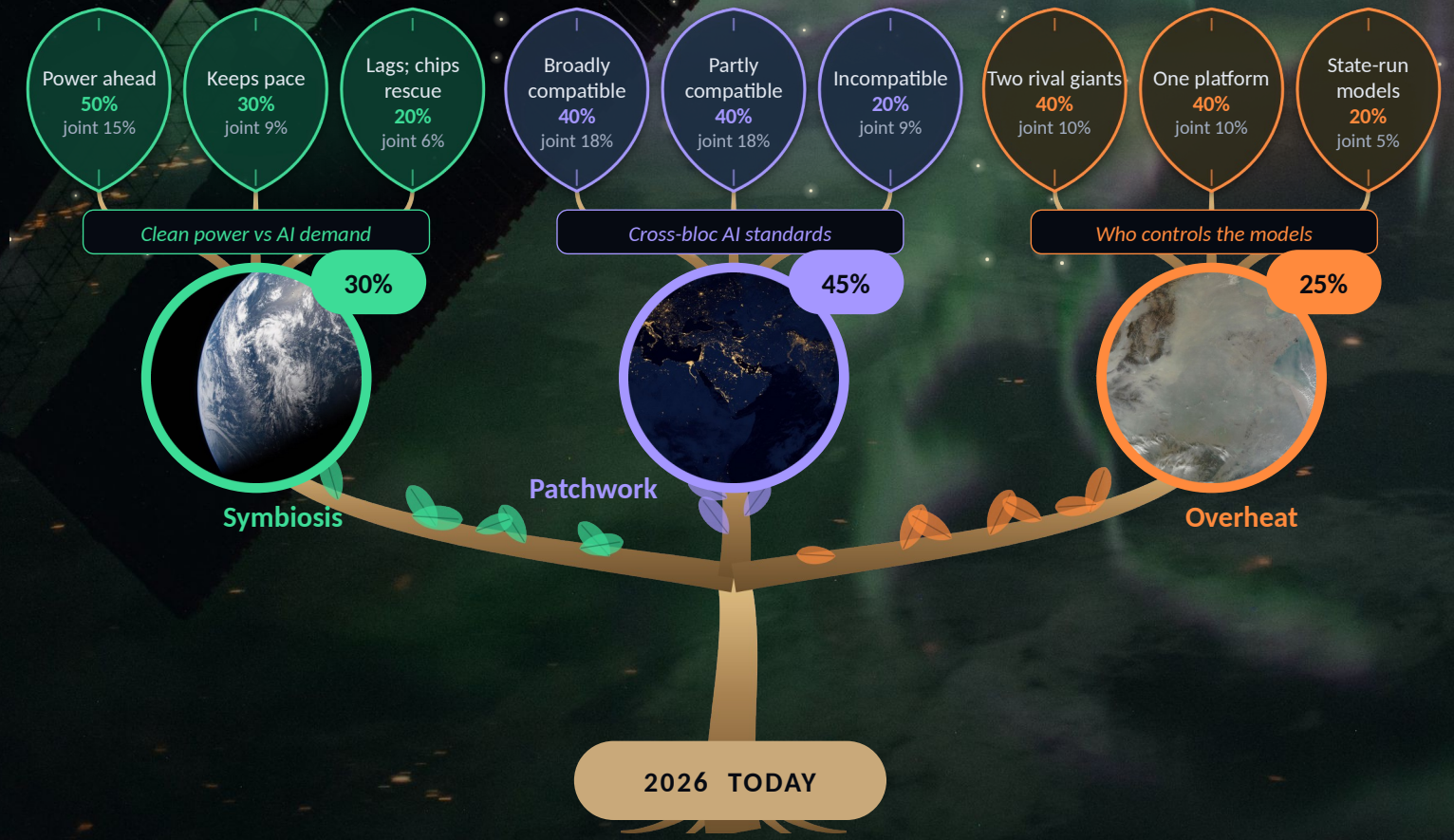
Design for nutrition, reliable supply and the crew's experience.



And gold? Still useful: it coats the mirrors of the Webb space telescope. But in 2041 it no longer measures wealth.

Which branch are we on?

Three worlds with planning weights, then one key driver per world split into three exclusive outcomes. Conditional odds sum to 100% per world, and the joint odds of all nine outcomes sum to 100%. Illustrative judgements, not calibrated forecasts.



How to read the tree

Level 1: worlds

Planning weights for the three worlds: 30%, 45%, 25%. They sum to 100%.

Level 2: one driver each

Three mutually exclusive outcomes per world, with conditional odds that sum to 100% within that world.

Joint odds

World x outcome. Worked example: Patchwork and partly compatible = 45% x 40% = 18%. All nine joint odds total 100%.

Why Patchwork leads

AI rules are already splitting by region; the clean-power race is not yet won.

Status

Illustrative judgements to support discussion, not calibrated forecasts.

A light only matters if it changes a decision

Each signpost has a record, a review trigger and a response. Triggers are proposed review rules measured against a pilot baseline, not external benchmarks.

SIGNPOST	WHAT WE RECORD	REVIEW TRIGGER	RESPONSE	POINTS TO
Clean power for AI	Clean energy built per new data centre; energy used per completed task	Two monthly reviews worse than baseline	**Switch to a smaller model or a manual route**	Overheat
Evidence quality	Share of outputs with a traceable source; material errors	Any unmitigated material error	**Pause automation; require human approval**	Symbiosis or Overheat
Portability	Time and failures when moving one test case between providers	A second provider fails the same test	**Add an adapter; avoid single-provider lock-in**	Patchwork
Orbital access	Debris-removal missions flown; launch delays	Removal still experimental by 2030	**Plan ground fallbacks for satellite services**	Overheat
Human agency	Can users understand, challenge and reverse a result?	A high-impact decision has no appeal route	**Redesign the handover before scaling**	All three worlds

Cadence. The pilot lead records weekly, the client owner reviews monthly, and the team revisits the scenario assumptions every quarter.

Sell better decisions, not more reports

Three proposed offers, one per world, share one reusable asset: a Decision Passport that records why a client should act, who owns the decision and what happens if it fails.

Future Design Lab

CLIENT Strategy and innovation leaders

Rehearse a new service in a digital twin before committing.

VALUE TEST Time to a testable decision

Verified Decisions and Continuity

CLIENT Operations and risk leaders

Check evidence, rehearse an outage and prove a fallback.

VALUE TEST Material errors; fallbacks completed

Interoperability Studio

CLIENT Product and technology leaders

Separate the core service from provider-specific adapters.

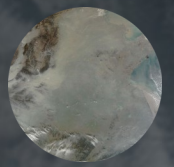
VALUE TEST Successful transfers; switching effort

DECISION PASSPORT | WORKED EXAMPLE: OMAR'S DISPATCH IN OVERHEAT (ILLUSTRATIVE REHEARSAL)

DECISION	ASSUMPTION	EVIDENCE	OWNER	REVIEW SIGNAL	FALLBACK
May the AI dispatch this van, or must Omar use the verified manual route?	Route and charging data are current enough for this dispatch.	Time-stamped sources, a human route check and pilot results, still to be collected.	Omar authorises; dispatchers can flag an alert.	Conflicting or untraceable alerts, or no charging available.	Verify by phone, use the manual route and tell the customer.

Proposed service concepts and an illustrative rehearsal, not existing products or results.

One causal story inside one world



The worlds branch, but a story inside a world must follow cause and effect. All eight Story Spine steps for Omar in Overheat, a fictional rehearsal.

01 ONCE UPON A TIME

Omar ran a depot whose customers depended on daily deliveries.

02 EVERY DAY

His team dispatched the vans on the AI's recommendation.

03 BUT ONE DAY

Two alerts conflicted, and the planned van could not charge.

04 BECAUSE OF THAT

Omar paused the recommendation before sending a driver.

05 BECAUSE OF THAT

He checked the records and called the depot to verify the route.

06 BECAUSE OF THAT

He used the manual fallback and logged why the decision changed.

07 UNTIL FINALLY

A verified alternative kept the delivery moving.

08 AND EVER SINCE THEN

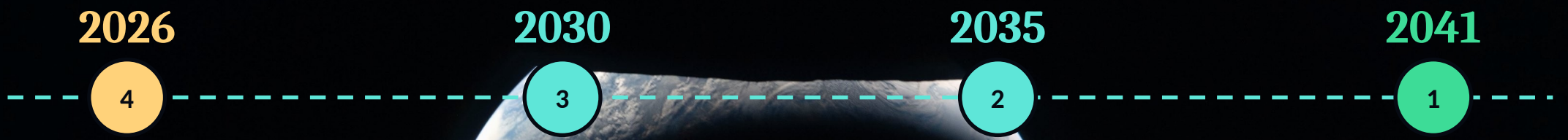
The team reviews every exception with a Decision Passport.

Takeaway (5 Cs carry-out). The cure is not perfect AI. It is a process that keeps evidence, ownership and a fallback visible.

From 2041 back to Monday morning

How the worlds become action. We start at the end in 2041 and plan step by step, right to left, so every move made today is useful in any world.

START HERE



No-regret moves this year

A digital-twin pilot with one client. A signpost dashboard that tracks the early-warning lights. A quarterly "day in 2041" hard-empathy workshop with each client team.

Energy-aware AI

Every client plans AI around its power footprint, and reskilling covers the 39% of skills expected to change.

Machine customers, verified supply

Each client runs a strategy for AI agents as buyers and a verified AI supply chain.

Press release from the future

"E,O&T clients thrive in whichever world arrived." Trusted AI, clean power and a usable orbit.

Creative fuel for the team. A shared commonplace book captures every signal and slow hunch; a planned stimuli queue (AI 2041, Imaginable, one sci-fi film a month) feeds the next round of scenarios.

Prove it is useful before scaling it

One client, one workflow, one limited rehearsal. All activities and gates below are proposed; no client commitment or benefit is claimed.

DAYS 1 TO 30

Define and observe

OWNERS Client owner and consultant

Map one workflow, record a manual baseline and interview five users.

GATE: agree success, error severity, data permissions and limits.

DAYS 31 TO 60

Prototype and rehearse

OWNERS Technical lead and client users

Build a simulation and a Decision Passport; rehearse all three worlds.

GATE: five users can name the evidence, owner and fallback.

DAYS 61 TO 90

Stress-test and decide

OWNERS Sponsor and independent reviewer

Run ten scripted cases: low power, false input, provider outage and normal operation.

GATE: every critical fallback works; no unresolved material error.

Continue, change or stop. Scale only when the pilot shows client-valued benefit and acceptable resource use.

Pilot limit. Five users and ten cases support learning, not proof of general reliability.

Rehearse, reflect, feed the next idea

A proposed monthly workshop that links the creative conditions, motivating feedback and future-thinking tools from this briefing.

01 GENERATE | 10 MIN

A safe room for alternatives

PSYCHOLOGICAL SAFETY

Write ideas silently, then share. The leader invites disagreement; challenge the assumption, not the person.

02 REHEARSE | 20 MIN

Learn through the scenario

OCTALYSIS, USED ETHICALLY

Pick a world and a role. A shared progress bar tracks evidence, owner and fallback; retries are allowed after feedback.

03 REFLECT | 15 MIN

Keep the useful surprise

AI AND SCIENCE FICTION

Try one strange AI or sci-fi variation as an idea, not a fact. Check its assumptions and keep a testable version in the commonplace book.

Ethical design. Reward learning and feedback, not speed or a public rank. No penalties for failed ideas.

Monthly stimuli queue. One cross-industry source, one science fiction story and one observation walk; revisit slow hunches.

CLOSING

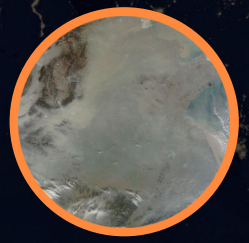
STORY SPINE **Carry-out message**

The Moon is closer than ever. So is 2041.

Three worlds, one discipline: make the assumptions visible, test the response, and keep a human decision owner in the loop.

Symbiosis
Build it

Patchwork
Shape it



Overheat
Insure against it

We do not predict the future. We prepare for all three, and we work every day for the first one.

Mohammad, Senior Futures Consultant, Em, Oli & Timmonds

NEXT Part 2, my personal reflection on the tools behind this briefing.

What I am taking with me

The tools that resonated, the change in my thinking, and why it matters in my professional life.

Three ideas from this course changed how I think about planning: possibility trees, working backwards, and counterfactual thinking. Together they gave me a structured language for something my work has always demanded: decisions about a future nobody can see.

In twenty years of network engineering and digital transformation, I have learned that a platform chosen today may still shape a business in ten years. Building the three worlds of 2041 for this assignment showed me a different way to approach that kind of decision. Drawing the possibility tree forced me to give each branch a probability, and the less likely branches raised questions a single forecast would never have asked. Working backwards from a 2041 press release, the way Amazon writes the press release before the product exists, made the first steps for 2026 obvious. Counterfactual thinking kept the exercise honest: writing Overheat next to Symbiosis stopped me from falling in love with the optimistic future.

What resonated most is that creativity here is not a talent but a structured skill, wild ideas plus strategy.

20 years

in networks and digital transformation, where decisions have long consequences

How I will put it into practice

A realistic timetable, a review loop and a personal routine that continues beyond this course.

My plan is concrete. From November, every new technical proposal I write will open with a one-page working backwards brief describing the result for the customer in three years, the assumptions behind it and the first test. Once a quarter, starting in January 2027, I will build a possibility tree for one roadmap decision, with three branches, nine sub-branches and probabilities, and review it three months later to see which branch reality followed. Before each major design review I will run one upward and one downward counterfactual with the team and record the risks it surfaces.

Personally, I live between Riyadh and Stockholm, and I will turn those flights into fuzzy focus time: no screens, just my stimuli queue and a commonplace book, so slow hunches have somewhere to land.

I will measure success simply: by June 2027, I will have built two possibility trees, completed the first three-month review, and recorded whether the findings changed a roadmap decision.

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Image credits

Slide 1: "A Breathtaking Earthset from Orion", Artemis II crew, 6 April 2026. NASA (art002e021278).

Slide 3: Night Earth observation from the ISS, Expedition 40. NASA (iss040e085126).

Slides 2, 4 and 14: "Home in the Distance", Artemis II. NASA (art002e020922).

Slide 5: "Earth in View", Artemis II (art002e023506); CADRE rovers concept, NASA/JPL-Caltech (PIA26296).

Slide 6: "Haze over eastern China", NASA Goddard (GSFC_20171208_Archive_e001269); ISS Expedition 35 (iss035e017673).

Slides 7, 10 and 17: "Black Marble: Africa, Europe, and the Middle East", NASA Goddard (GSFC_20171208_Archive_e001591).

Slides 5 to 7 and 16: NASA VR training (jsc2026e000427), Advanced Air Mobility concepts (AAM-Concept-2021, automation concept), Mission Control (jsc2026e019925), Pleiades supercomputer (ARC-2008-ACD08-0272-005).

Slide 8: NASA drone flight test (AFRC2026-0152-22); Cygnus in Canadarm2 (iss071e418235); Expedition 66 chile peppers (iss066e023179).

Slide 9: ISS water recovery (iss072e143163); sand dunes and irrigation in Saudi Arabia (PIA11069); Earth limb and Moon (iss051e039977); Veg-01 harvest (iss040e009126); JWST golden mirror (GSFC_20171208_Archive_e000368).

World thumbnails on slides 2, 4, 10, 12, 13 and 17 reuse the images of slides 5, 6 and 7.

All NASA images via the NASA Image and Video Library, https://images.nasa.gov (public domain).

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AI DECLARATION

Use of AI

I acknowledge the use of Claude by Anthropic (https://claude.ai) and ChatGPT by OpenAI (https://chatgpt.com) to help generate and develop scenario ideas and variants, critique and develop wording, framework applications, scenario logic and probabilities, design the Decision Passport, pilot and workshop, find and check sources, and arrange and format the slides. The prompts used include "Propose solution directions for each company and the course concepts they would apply", "Draft three different future scenarios for 2041 with personas, built on the course tools", "Check every slide against the assignment requirements and rubric", "Merge the strongest ideas from my two drafts into one deck" and "Build the deck as a PowerPoint with my photo and a dark cinematic layout". The output was used as a starting point that I selected, reviewed, edited and verified; the choice of company and direction, the final scenarios, and the reflection based on my own experience and plans are mine, and I remain responsible for the submission.